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Complex frequency-shifted multi-axial perfectly matched layer for elastic wave modelling on curvilinear grids

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SUMMARY

Perfectly matched layer (PML) is an efficient absorbing technique for numerical wave simulations. Since it appeared, various improvements have been made. The complex frequency-shifted PML (CFS-PML) improves the absorbing performance for near-grazing incident waves and evanescent waves. The auxiliary differential equation (ADE) formulation of the PML provides a convenient unsplit-field PML implementation that can be directly used with high order time marching schemes. The multi-axial PML (MPML) stabilizes the PML on anisotropic media. However, these improvements were generally developed for Cartesian grids. In this paper, we extend the ADE CFS-PML to general curvilinear (non-orthogonal) grids for elastic wave modelling. Unlike the common implementations to absorb the waves in the computational space, we apply the damping along the perpendicular direction of the PML layer in the local Cartesian coordinates. Further, we relate the perpendicular and parallel components of the gradient operator in the local Cartesian coordinates to the derivatives in the curvilinear coordinates, to avoid mapping the wavefield to the local Cartesian coordinates. It is thus easy to be incorporated with numerical schemes on curvilinear grids. We derive the PML equations for the interior region and for the free surface separately because the free surface boundary condition modifies the elastic wave equations. We show that the elastic wave modelling on curvilinear grids exhibits anisotropic effects in the computational space, which may lead to unstable simulations. To stabilize the simulation, we adapt the MPML strategy to also absorb the wavefield along the two parallel directions of the PML. We illustrate the stability of this ADE CFS-MPML for finite-difference elastic wave simulations on curvilinear grids by two numerical experiments.

Key words: Numerical solutions; Earthquake ground motions; Computational seismology; Theoretical seismology; Wave propagation.

1 INTRODUCTION

The perfectly matched layer (PML; Bérenger 1994) is an efficient technique to absorb the outgoing waves surrounding the computational domain for numerical wave modelling. The original PML proposed by Bérenger (1994) was developed for Maxwell's equations and was implemented in split-field formulation by non-physically splitting the wavefield components and equations. This split-field PML was shortly extended for elastic wave equations due to its superior performance (e.g. Chew & Liu 1996; Hastings *et al.* 1996; Collino & Tsogka 2001; Festa & Nielsen 2003; Komatitsch & Tromp 2003; Marcinkovich & Olsen 2003). The PML can also be interpreted and constructed based upon the concept of complex coordinate stretching (analytic continuation; e.g. Chew & Weedon 1994; Teixeira & Chew 2000), artificial anisotropic medium (uniaxial PML or UPML; e.g. Sacks *et al.* 1995; Gedney 1996), or modified modal solution technique (modal PML; e.g. Hagstrom 2003; Appelö & Kreiss 2006). These alternative interpretations have lead to the development of different unsplit-field implementation of the PML (e.g. Zheng & Huang 2002; Ramadan 2003; Wang & Tang 2003; Zeng & Liu 2004; Appelö & Kreiss 2006).

Kuzuoglu & Mittra (1996) proposed the complex frequency-shifted PML (CFS-PML) using a frequency dependent stretching function. The CFS-PML was found to be important to absorb evanescent waves (Bérenger 2002; Festa *et al.* 2005) and near-grazing incident waves (Drossaert & Giannopoulos 2007b; Komatitsch & Martin 2007; Zhang & Shen 2010). The CFS-PML was first implemented in a split-field formulation (Gedney 1998). Later, different unsplit-field implementations of the CFS-PML were developed by using convolution terms

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(C-PML; e.g. Roden & Gedney 2000; Wang & Tang 2003; Drossaert & Giannopoulos 2007a; Komatitsch & Martin 2007), integral terms (e.g. Zeng & Liu 2004; Drossaert & Giannopoulos 2007b), matched Z-transform (MZT CFS-PML;e.g. Shi *et al.* 2012), or auxiliary differential equations (ADE CFS-PML; e.g. Wang & Liang 2006; Kristek *et al.* 2009; Qin *et al.* 2009; Gedney & Zhao 2010; Martin *et al.* 2010; Zhang & Shen 2010; Duru & Kreiss 2012; Feng & Li 2013). The ADE CFS-PML results in a complete first-order partial differential equations and can be solved by the same numerical scheme for the physical elastic wave equations even an high order time marching scheme is used.

In aforementioned PMLs, the wavefield is only damped along the perpendicular direction of the PML. To overcome the unstable problem of the split-field PML in some anisotropic media, Meza-Fajardo & Papageorgiou (2008) proposed the multi-axial PML (MPML) that also damps the wavefield along the other two directions parallel to the PML. They showed that the MPML can stabilize the numerical simulation for any model of anisotropic elastic media because the stability is related to the ratio of the damping profiles.

The computational domains of numerical wave simulations are not always rectangular. When the grids or elements inside the PML are not rectangular, a curvilinear coordinate system is needed. Some PMLs have been proposed for spherical or cylindrical coordinates that based upon the concept of the complex coordinate stretching (e.g. Teixeira & Chew 1997a,b; Collino & Monk 1998; Liu 1999) or the artificial anisotropic medium (e.g. Zheng & Huang 2002). The conformal PML for orthogonal curvilinear coordinates was also developed from the change on the space metric (e.g. Teixeira & Chew 1998, 2000; Festa & Vilotte 2005). The resulting PML equations are complicated and we have not seen any effort to use CFS-PML in these curvilinear coordinates. Unstable simulations was also found for the PML on curvilinear grids (Teixeira *et al.* 2001). The situation is more complicated for the PML on general curvilinear (non-orthogonal) coordinates. There are only several attempts to develop the PML for non-orthogonal grids by solving the covariant and contravaiant components of the wavefield in the curvilinear coordinates and absorbing the wavefield along the curvilinear coordinates (e.g. Navarro *et al.* 1994; Roden & Gedney 1997; Oliveira & Sobrinho 2007; Rao & Wang 2013). Such approaches are equivalent to apply the PML in the Cartesian coordinates in the computational space along the direction perpendicular to the PML, but that direction in the physical space is not perpendicular to the PML due to the non-orthogonal grids. Gao & Zhang (2008) and Zhang & Gao (2011) proposed another strategy to implement the PML for irregular absorbing layers in a local Cartesian coordinate system perpendicular to the PML. Their approach provides an convenient way to implement the PML on curvilinear grids.

In this paper, we extend the ADE CFS-PML to general curvilinear (non-orthogonal) grids for linearized elastic wave modelling. Unlike the common PML implementations for curvilinear coordinates absorbing the waves in the computational space, we apply the damping along the perpendicular direction of the PML layer in the local Cartesian coordinate system. The similar approach has been used by Gao & Zhang (2008) and Zhang & Gao (2011) for irregular PML layers using the split-field PML. Their approach requires all the variables are defined in the local Cartesian coordinate system. In our implementation, instead of mapping all the variables to the local Cartesian coordinate system, we relate the perpendicular and parallel components of the gradient operator in Cartesian coordinates to the derivatives in the curvilinear coordinates. The MPML scheme proposed by Meza-Fajardo & Papageorgiou (2008) are adapted to the ADE CFS-PML scheme Zhang & Shen (2010) in the curvilinear coordinates (ADE CFS-MPML). We illustrate the stability of this ADE CFS-MPML for finite-difference elastic wave simulations on curvilinear grids by two numerical experiments.

2 ELASTIC WAVE EQUATION IN CURVILINEAR COORDINATES AND STABILITY OF PML

Given a linearly elastic medium, the momentum equation

$$\rho \mathbf{v}_{,t} = \nabla \cdot \boldsymbol{\sigma} + \mathbf{f}, \tag{1}$$

the time derivative of the stress–strain relation (constitutive relation, Hooke's law)

$$\boldsymbol{\sigma}_{,t} = \boldsymbol{c} : \boldsymbol{\varepsilon}_{,t}, \tag{2}$$

the time derivative of the infinitesimal strain tensor

$$\boldsymbol{\varepsilon}_{,t} = \frac{1}{2}\left[\nabla \mathbf{v} + (\nabla \mathbf{v})^T\right], \tag{3}$$

and initial boundary conditions completely govern the propagation of the elastic seismic wave generated by an external force or internal earthquake $\mathbf{f}$. In the above equations, t is time, ρ is density, $\mathbf{v}$ is the velocity vector, $\boldsymbol{\sigma}$ and $\boldsymbol{\varepsilon}$ are the stress and strain tensor, respectively, and matrix $\mathbf{c}$ is the fourth order stiffness tensor. A comma followed by t denotes the first-order derivation with respect to time. Eqs (1)–(3) can be written as a first order hyperbolic system in terms of the velocity and stress

$$\frac{\partial \mathbf{U}}{\partial t} = \mathbf{A}\frac{\partial \mathbf{U}}{\partial x} + \mathbf{B}\frac{\partial \mathbf{U}}{\partial y} + \mathbf{C}\frac{\partial \mathbf{U}}{\partial z} + \mathbf{F}, \tag{4}$$

where $\mathbf{U}$ is the velocity–stress vector

$$\mathbf{U} = (v_x, v_y, v_z, \sigma_{xx}, \sigma_{yy}, \sigma_{zz}, \sigma_{yz}, \sigma_{xz}, \sigma_{xy})^T, \tag{5}$$

$\mathbf{F}$ is the source vector

$$\mathbf{F} = (\, f_x/\rho,\ f_y/\rho,\ f_z/\rho,\ -\dot{M}_{xx},\ -\dot{M}_{yy},\ -\dot{M}_{zz},\ -\dot{M}_{yz},\ -\dot{M}_{xz},\ -\dot{M}_{xy}\,)^T, \tag{6}$$

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and coefficient matrices

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 0 & 1/\rho & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/\rho \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/\rho & 0 \\ \lambda+2\mu & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \lambda & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \lambda & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \mu & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \mu & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \tag{7}$$

B and **C** are similar and can be easily derived.

To simulate seismic wave propagation in the presence of surface topography, a curvilinear grid should be used (e.g. Zhang & Chen 2006; Zhang *et al.* 2012). We can map a curvilinear grid in physical space (x, y, z) to a uniform rectangular grid in computational space (ξ, η, ζ) through coordinate transformation

$$x = x(\xi, \eta, \zeta), \qquad y = y(\xi, \eta, \zeta), \qquad z = z(\xi, \eta, \zeta). \tag{8}$$

Consequently, the velocity–stress equation (eq. 4) in Cartesian coordinates is also transformed to this computational space using chain rules,

$$\frac{\partial \mathbf{U}}{\partial t} = \mathbf{A}'\frac{\partial \mathbf{U}}{\partial \xi} + \mathbf{B}'\frac{\partial \mathbf{U}}{\partial \eta} + \mathbf{C}'\frac{\partial \mathbf{U}}{\partial \zeta} + \mathbf{F}, \tag{9}$$

where

$$\mathbf{A}' = \begin{pmatrix} 0 & 0 & 0 & \xi_{,x}/\rho & 0 & 0 & 0 & \xi_{,z}/\rho & \xi_{,y}/\rho \\ 0 & 0 & 0 & 0 & \xi_{,y}/\rho & 0 & \xi_{,z}/\rho & 0 & \xi_{,x}/\rho \\ 0 & 0 & 0 & 0 & 0 & \xi_{,z}/\rho & \xi_{,y}/\rho & \xi_{,x}/\rho & 0 \\ \xi_{,x}(\lambda+2\mu) & \xi_{,y}\lambda & \xi_{,z}\lambda & 0 & 0 & 0 & 0 & 0 & 0 \\ \xi_{,x}\lambda & \xi_{,y}(\lambda+2\mu) & \xi_{,z}\lambda & 0 & 0 & 0 & 0 & 0 & 0 \\ \xi_{,x}\lambda & \xi_{,y}\lambda & \xi_{,z}(\lambda+2\mu) & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \xi_{,z}\mu & \xi_{,y}\mu & 0 & 0 & 0 & 0 & 0 & 0 \\ \xi_{,z}\mu & 0 & \xi_{,x}\mu & 0 & 0 & 0 & 0 & 0 & 0 \\ \xi_{,y}\mu & \xi_{,x}\mu & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \tag{10}$$

$\mathbf{B}'$ and $\mathbf{C}'$ are similar and omitted here.

Though the original wave equation in Cartesian coordinates (eq. 4) is isotropic, it exhibits anisotropy in curvilinear coordinates (eq. 9) due to coordinate transformation if we observe the wavefield in the computational space, where the wave equation is numerically solved. This anisotropic effect can be easily understood from a geometrical viewpoint. Fig. 1(a) illustrates a deformed grid with 55° angle between grid lines. The wave front from a point source shows a spherical shape on it. In the computational space, the wave front is not spherical but exhibits different speeds at different directions (Fig. 1b), which has the similar characters as the wave propagation in an anisotropic medium. Halpern *et al.* (2011) have shown that the PML in some anisotropic media may become amplifying instead of damping. Bécache *et al.* (2003) derived a geometrical stability condition for stable PML solution in anisotropic media: in the slowness diagram, the slowness vector and the group velocity should be oriented in the same direction with respect to coordinate axes. However, for the slowness diagram in the curvilinear coordinates as shown in Fig. 1(c), the slowness vector (arrow with dashed line) and the group velocity vector (arrow with solid line) are at the opposite directions of the horizontal axis at directions denoted by the red lines, and are at the opposite directions of the vertical axis at directions indicated by the thick black lines. That means the PML at these directions becomes amplifying layer. In following sections, we will adapt the multi-axial PML for anisotropic media (Meza-Fajardo & Papageorgiou 2008) to stabilize the PML solution on curvilinear grids.

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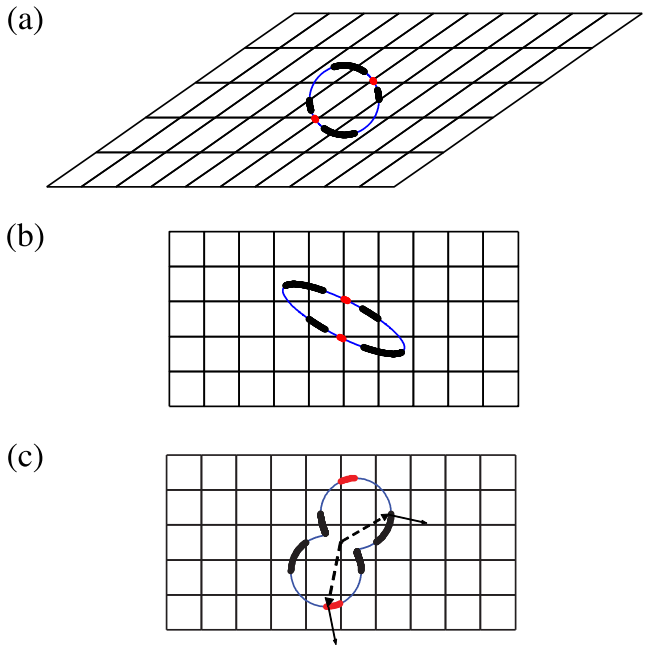


Figure 1. Geometric illustration of stable and unstable portions of the wave front and slowness curve on curvilinear grids according to the geometric stability criterion in Bécache *et al.* (2003). Panel (a) shows the deformed grid (35° angle between the grid lines) and the wave front (blue line) from a point source in the physical space. Panel (b) depicts the corresponding rectangular grid in the computational space and the deformed wave front (blue line). Panel (c) is the slowness diagram (blue line) in the curvilinear coordinates in the computational space. The thick red and black lines illustrate the directions at which the vertical and horizontal PMLs become amplifying layers, respectively. Arrows with dashed and solid line in the slowness diagram represent the slowness vector and group velocity vector, respectively.

3 ADE CFS-MPML AT NON-FREE-SURFACE GRIDPOINTS ON CURVILINEAR GRIDS

PML can be interpreted as a complex coordinate stretching (Chew & Weedon 1994), which replaces the spatial coordinate x with a complex stretched coordinate $\tilde{x}$ for a PML terminating the positive x-axis from $x = 0$,

$$\tilde{x} = \int_0^x s_x(x')\,\mathrm{d}x', \tag{11}$$

where s_x is the complex stretching function that determines the characteristics of the PML model. The complex stretching function of the original PML (Bérenger 1994) has the form,

$$s_x(x) = 1 + \frac{d_x(x)}{\mathrm{i}\omega}, \tag{12}$$

while the CFS-PML (Kuzuoglu & Mittra 1996) introduces another two factors to ensure the PML causality,

$$s_x(x) = \beta_x(x) + \frac{d_x(x)}{\alpha_x(x) + \mathrm{i}\omega}, \tag{13}$$

where $d_x \geq 0$ is the attenuation factor, $\alpha_x \geq 0$ is the frequency shifted factor and $\beta_x \geq 1$ is the scaling factor. Generally, the PML or the CFS-PML only applies the above coordinate stretching (eq. 11) to the coordinate perpendicular to the PML layer, while leaves the other coordinates unchanged to achieve the perfectly matched property cross the interface of the PML and the physical domain. The MPML proposed by Meza-Fajardo & Papageorgiou (2008) applies the coordinate stretching along all the three directions to overcome the unstable problem of the split-field PML in anisotropic media.

In this section, we derive the ADE CFS-MPML equations for the elastic velocity–stress equations in the curvilinear coordinates. We also derive the ADE CFS-MPML equations at the free surface to take into account the free surface boundary condition in the next section.

3.1 Gradient decomposition in local Cartesian coordinates

To derive the PML equations by damping the wavefield along the three directions of the local Cartesian coordinates, we need to decompose the gradient operator onto these three directions.

In the following, We denote the covariant axis (ξ, η, ζ) of the curvilinear grid as (ξ^1, ξ^2, ξ^3). The covariant base vectors $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ (Fig. 2) are the local directions of the covariant axis along the grid lines and can be calculated by

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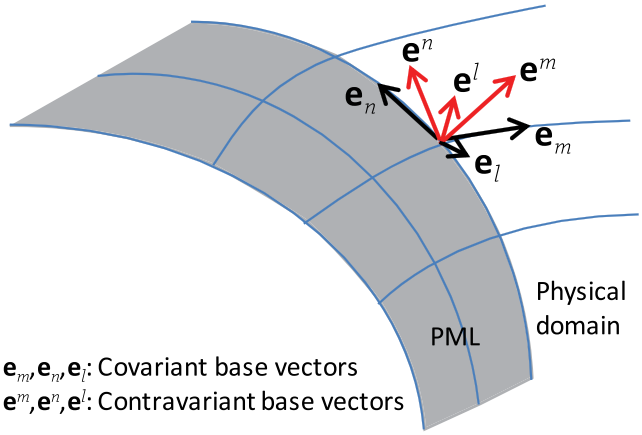


Figure 2. Covariant and contravariant base vectors on curvilinear grids. The covariant vectors (black vectors, $\mathbf{e}_m$, $\mathbf{e}_n$, $\mathbf{e}_l$) are parallel to grid lines, which are presented by blue lines. The contravariant vectors (red vectors, $\mathbf{e}^m$, $\mathbf{e}^n$, $\mathbf{e}^l$) are perpendicular to grid lines.

$$\mathbf{e}_n = \sum_{i=1}^{3} x^i_{,n}\mathbf{i} \qquad (n = 1, 2, 3), \tag{14}$$

where x^i is the ith axis of and $\mathbf{i}$ is the ith base vector of the global Cartesian coordinates (x, y, z). The n following a comma means the derivation with respect to the nth covariant axis of the curvilinear coordinates. The curvilinear coordinates also have another set base vectors called the contravariant base vectors (Fig. 2)

$$\mathbf{e}^m = \sum_{i=1}^{3} \xi^m_{,i}\mathbf{i} \qquad (m = 1, 2, 3). \tag{15}$$

The contravariant base vector $\mathbf{e}^m$ has the property that it is orthogonal to the other two covariant base vectors $\mathbf{e}_n$ and $\mathbf{e}_l$

$$\mathbf{e}^m \parallel (\mathbf{e}_n \times \mathbf{e}_l). \tag{16}$$

In the followings, we will use $m, n, l \in (1, 2, 3)$ and $i, j, k \in (1, 2, 3)$ to represent the subscripts for the curvilinear coordinates and Cartesian coordinates, respectively. The summation convention is not applied unless explicitly declared.

At a gridpoint inside the PML, we can define local Cartesian coordinates (x', y', z') such that the base vector $\mathbf{n}_1$ is perpendicular to the surface of the PML and coincides with the direction of $\mathbf{e}^m$, $\mathbf{n}_2$ is in the direction of $\mathbf{e}_n$ and $\mathbf{n}_3$ is orthogonal to $\mathbf{n}_1$ and $\mathbf{n}_2$ as

$$\mathbf{n}_1 = \frac{\mathbf{e}^m}{|\mathbf{e}^m|}, \qquad \mathbf{n}_2 = \frac{\mathbf{e}_n}{|\mathbf{e}_n|}, \qquad \mathbf{n}_3 = \mathbf{n}_1 \times \mathbf{n}_2, \tag{17}$$

where $\times$ means a cross product.

To implement the ADE CFS-MPML on this local Cartesian coordinates, we need to project the ∇ operator in the curvilinear coordinates onto each $\mathbf{n}_i$ direction as

$$\nabla = \nabla_1 + \nabla_2 + \nabla_3, \tag{18}$$

where ∇_1 is along $\mathbf{n}_1$, the major damping direction; ∇_2 and ∇_3 are along $\mathbf{n}_2$ and $\mathbf{n}_3$, the auxiliary damping directions. From the expression of ∇ in the curvilinear coordinates

$$\nabla = \sum_{n=1}^{3} \mathbf{e}^n \frac{\partial}{\partial \xi^n}, \tag{19}$$

we can get (Thompson *et al.* 1985, chap. 3, eq. 79)

$$\nabla_1 = \frac{\boldsymbol{e}^m}{|\boldsymbol{e}^m|}\left(\frac{\boldsymbol{e}^m}{|\boldsymbol{e}^m|}\cdot\nabla\right) = \frac{\boldsymbol{e}^m}{|\boldsymbol{e}^m|}\left(\frac{\boldsymbol{e}^m}{|\boldsymbol{e}^m|}\cdot\sum_{n=1}^{3}\boldsymbol{e}^n\frac{\partial}{\partial\xi^n}\right) = \boldsymbol{e}^m\left(\frac{1}{g^{mm}}\sum_{n=1}^{3} g^{mn}\frac{\partial}{\partial\xi^n}\right), \tag{20}$$

where

$$g^{mn} = \boldsymbol{e}^m \cdot \boldsymbol{e}^n = \sum_{i=1}^{3} \xi^m_{,i}\xi^n_{,i}, \tag{21}$$

and

$$\nabla_2 = \frac{\boldsymbol{e}_n}{|\boldsymbol{e}_n|}\left(\frac{\boldsymbol{e}_n}{|\boldsymbol{e}_n|}\cdot\sum_{m=1}^{3}\boldsymbol{e}^m\frac{\partial}{\partial\xi^m}\right) = \boldsymbol{e}_n\left(\frac{1}{g_{nn}}\sum_{m=1}^{3} g^m_n\frac{\partial}{\partial\xi^m}\right), \tag{22}$$

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where

$$g_n^m = \mathbf{e}^m \cdot \mathbf{e}_n = \delta_{mn} = \begin{cases} 1, & m = n \\ 0, & m \neq n \end{cases}, \tag{23}$$

and

$$\nabla_3 = \boldsymbol{n}_3 \left(\boldsymbol{n}_3 \cdot \sum_{m=1}^{3} \boldsymbol{e}^m \frac{\partial}{\partial \xi^m} \right) = \boldsymbol{n}_3 \left(\sum_{m=1}^{3} h_3^m \frac{\partial}{\partial \xi^m} \right), \tag{24}$$

where

$$h_3^m = \boldsymbol{n}_3 \cdot \boldsymbol{e}^m, \tag{25}$$

respectively.

With the decomposed operator ∇ (eq. 18), the PML scheme can be easily implemented in the direction we want to damp the seismic wave. As we use the first-order stress–velocity equations (eqs 1–3) to solve the elastodynamic problem, we interpret the ADE CFS-MPML scheme for the momentum equations and the Hooke's law separately.

3.2 ADE CFS-MPML for momentum equations

In the local Cartesian coordinates, we can write the ADE CFS-PML for the moment equation as

$$\rho \mathbf{v}_{,t} = \nabla \cdot \boldsymbol{\sigma} + \left(\frac{1}{\beta_m} - 1 \right) \nabla_\perp^m \cdot \boldsymbol{\sigma} - \frac{1}{\beta_m} \mathbf{V}, \tag{26}$$

where the memory variable $\mathbf{V}$ satisfies

$$\mathbf{V}_{,t} + \left(\alpha_m + \frac{d_m}{\beta_m} \right) \mathbf{V} = \frac{d_m}{\beta_m} \nabla_\perp^m \cdot \boldsymbol{\sigma}. \tag{27}$$

Since $\nabla_\perp^m = \nabla_1^m$ and $\mathbf{v}$, $\nabla \cdot \boldsymbol{\sigma}$ and $\mathbf{V}$ are invariant in different coordinates, we can directly solve the above equations in the curvilinear coordinates instead of rotating all the variables to the local Cartesian coordinates by replaced $\nabla_\perp^m$ with ∇_1^m

$$\rho \mathbf{v}_{,t} = \nabla \cdot \boldsymbol{\sigma} + \left(\frac{1}{\beta_m} - 1 \right) \nabla_1^m \cdot \boldsymbol{\sigma} - \frac{1}{\beta_m} \mathbf{V}, \tag{28}$$

and

$$\mathbf{V}_{,t} + \left(\alpha_m + \frac{d_m}{\beta_m} \right) \mathbf{V} = \frac{d_m}{\beta_m} \nabla_1^m \cdot \boldsymbol{\sigma}. \tag{29}$$

The above ADE CFS-PML only damps the wavefield along the perpendicular direction of the PML. The ADE CFS-MPML damps the wavefield along all the three directions of the local Cartesian coordinate system as

$$\rho \mathbf{v}_{,t} = \nabla \cdot \boldsymbol{\sigma} + \sum_{p=1}^{3} \left\{ \left(\frac{1}{\beta_p^m} - 1 \right) \nabla_p^m \cdot \boldsymbol{\sigma} - \frac{1}{\beta_p^m} \mathbf{V}^p \right\}, \tag{30}$$

where the pth memory variable $\mathbf{V}^p$ satisfies the following equation:

$$\mathbf{V}_{,t}^p + \left(\alpha_p^m + \frac{d_p^m}{\beta_p^m} \right) \mathbf{V}^p = \frac{d_p^m}{\beta_p^m} \nabla_p^m \cdot \boldsymbol{\sigma} \qquad (p = 1, 2, 3). \tag{31}$$

In these equations, for the main damping direction $\mathbf{n}_1$, the parameters α_1^m, β_1^m and d_1^m take the profiles and optimal values following Zhang & Shen (2010). For the auxiliary damping directions, we set $\alpha = 0$ and $\beta = 1$ to recover the standard PML, and set d_p^m following Meza-Fajardo & Papageorgiou (2008) as

$$d_2^m = p_{12} d_1^m, \quad d_3^m = p_{13} d_1^m, \tag{32}$$

in which the parameters p_{12} and p_{13} are between 0 and 1. At the corners where different PMLs overlap, we can use either PML to absorb the wavefield due to the multi-axial damping by adding the values of the damping profiles of one to the other. In this work, we use the PMLs terminating ξ-axis at the corners of the ξ- and η-axis, and use the PML of ζ-axis at the corners at the bottom of the computational domain. The three gradient terms in eqs (30) and (31) can be written as (utilizing eqs 20–24)

$$\nabla_1^m \cdot \boldsymbol{\sigma}|_i = \sum_{j=1}^{3} \left[\xi_{,j}^m \left(\frac{1}{g^{mm}} \sum_{n=1}^{3} g^{mn} \frac{\partial}{\partial \xi^n} \right) \sigma_{ij} \right] = \frac{1}{g^{mm}} \sum_{n=1}^{3} g^{mn} \sum_{j=1}^{3} \left(\xi_{,j}^m \frac{\partial \sigma_{ij}}{\partial \xi^n} \right) = \frac{1}{g^{mm}} \sum_{n=1}^{3} g^{mn} \left(\xi_{,x}^m \frac{\partial \sigma_{ix}}{\partial \xi^n} + \xi_{,y}^m \frac{\partial \sigma_{iy}}{\partial \xi^n} + \xi_{,z}^m \frac{\partial \sigma_{iz}}{\partial \xi^n} \right), \tag{33}$$

$$\nabla_2^m \cdot \boldsymbol{\sigma}|_i = \sum_{k=1}^{3} \left[x_{,n}^k \left(\frac{1}{g_{nn}} \sum_{m=1}^{3} g_n^m \frac{\partial}{\partial \xi^m} \right) \sigma_{ik} \right] = \frac{1}{g_{nn}} \sum_{m=1}^{3} g_n^m \sum_{k=1}^{3} \left(x_{,n}^k \frac{\partial \sigma_{ik}}{\partial \xi^m} \right), \tag{34}$$

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and

$$\nabla_3^m \cdot \boldsymbol{\sigma}|_i = \sum_{j=1}^{3} n_3^j \sum_{m=1}^{3} h_3^m \frac{\partial \sigma_{ij}}{\partial \xi^m}. \tag{35}$$

3.3 ADE CFS-MPML for Hooke's law equation

Similar, the multi-axial ADE CFS-PML equations of the Hooke's law are

$$\varepsilon_{ij,t} = \frac{1}{2}\left[\nabla \mathbf{v} + (\nabla \mathbf{v})^T\right]\big|_{ij} + \sum_{p=1}^{3}\left\{\left(\frac{1}{\beta_p^m} - 1\right)\frac{1}{2}\left[\nabla_p^m \mathbf{v} + \left(\nabla_p^m \mathbf{v}\right)^T\right]\big|_{ij} - \frac{1}{\beta_p^m} E_{ij}^p\right\}, \tag{36}$$

and

$$E_{ij,t}^p + \left(\alpha_p^m + \frac{d_p^m}{\beta_p^m}\right) E_{ij}^p = \frac{d_p^m}{\beta_p^m}\frac{1}{2}\left[\nabla_p^m \mathbf{v} + \left(\nabla_p^m \mathbf{v}\right)^T\right]\big|_{ij}. \tag{37}$$

From eqs (20), (22) and (24), the derivative terms in above equations are

$$\left[\nabla_1^m \mathbf{v} + \left(\nabla_1^m \mathbf{v}\right)^T\right]\big|_{ij} = \frac{\xi_{,i}^m}{g^{mm}}\left(g^{m1}\frac{\partial v_j}{\partial \xi^1} + g^{m2}\frac{\partial v_j}{\partial \xi^2} + g^{m3}\frac{\partial v_j}{\partial \xi^3}\right) + \frac{\xi_{,j}^m}{g^{mm}}\left(g^{m1}\frac{\partial v_i}{\partial \xi^1} + g^{m2}\frac{\partial v_i}{\partial \xi^2} + g^{m3}\frac{\partial v_i}{\partial \xi^3}\right), \tag{38}$$

$$\left[\left(\nabla_2^m \mathbf{v}\right) + \left(\nabla_2^m \mathbf{v}\right)^T\right]\big|_{ij} = \frac{x_{,n}^i}{g_{nn}}\left(g_n^1\frac{\partial v_j}{\partial \xi^1} + g_n^2\frac{\partial v_j}{\partial \xi^2} + g_n^3\frac{\partial v_j}{\partial \xi^3}\right) + \frac{x_{,n}^j}{g_{nn}}\left(g_n^1\frac{\partial v_i}{\partial \xi^1} + g_n^2\frac{\partial v_i}{\partial \xi^2} + g_n^3\frac{\partial v_i}{\partial \xi^3}\right), \tag{39}$$

and

$$\left[\left(\nabla_3^m \mathbf{v}\right) + \left(\nabla_3^m \mathbf{v}\right)^T\right]\big|_{ij} = n_3^i \sum_{m=1}^{3} h_3^m \frac{\partial v_j}{\partial \xi^m} + n_3^j \sum_{m=1}^{3} h_3^m \frac{\partial v_i}{\partial \xi^m}. \tag{40}$$

We note that the ADE CFS-MPML eqs (30), (31), (36) and (37) on the curvilinear coordinates are expressed as a correction to the original wave equations (eqs 1 and 3), similar to the ADE CFS-PML in the Cartesian coordinates (Zhang & Shen 2010). They can be easily incorporated in an existing code.

4 ADE CFS-MPML AT THE FREE SURFACE ON CURVILINEAR GRIDS

The previous section discusses the basic equations of PML in the general curvilinear coordinates and ignores the existence of the free surface, where special mathematic equations are utilized in CG-FDM scheme (Zhang & Chen 2006; Zhang *et al.* 2012) according to the free-surface boundary condition. In this section, we discuss the implementation of the free-surface condition in the PML when the absorbing layer reaches the free surface.

4.1 ADE CFS-MPML for the moment equation at the free surface

The free-surface boundary condition is that the traction equals zero:

$$\boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{0}, \tag{41}$$

To apply traction imaging to the ghost points above the free surface (Zhang & Chen 2006; Zhang *et al.* 2012), we use the following conservative form of ∇ (Thompson *et al.* 1985, , chap. 3, eq. 43) in eqs (30) and (31):

$$\nabla \cdot \boldsymbol{\sigma} = \frac{1}{J}\sum_{n=1}^{3}\frac{\partial (J \boldsymbol{e}^n \cdot \sigma)}{\partial \xi^n}, \tag{42}$$

where J is the Jacobian. The corresponding gradient decomposition becomes

$$\nabla_1^m \cdot \boldsymbol{\sigma} = \frac{\mathbf{e}^m}{|\mathbf{e}^m|}\left[\frac{\mathbf{e}^m}{|\mathbf{e}^m|} \cdot \frac{1}{J}\sum_{n=1}^{3}\frac{\partial (J \boldsymbol{e}^n \cdot \sigma)}{\partial \xi^n}\right] = \frac{\mathbf{e}^m}{g^{mm} J}\sum_{i=1}^{3}\left[\xi_{,i}^m \sum_{n=1}^{3}\frac{\partial (J \sum_{j=1}^{3}(\xi_{,j}^n \sigma_{ij}))}{\partial \xi^n}\right]. \tag{43}$$

By writing components and considering the traction definitions, we obtain the following:

$$\nabla_1^m \cdot \boldsymbol{\sigma}|_k = \frac{\xi_{,k}^m}{g^{mm} J}\left[\xi_{,x}^m\left(\frac{\partial Txx}{\partial \xi} + \frac{\partial Txy}{\partial \eta} + \frac{\partial Txz}{\partial \zeta}\right) + \xi_{,y}^m\left(\frac{\partial Tyx}{\partial \xi} + \frac{\partial Tyy}{\partial \eta} + \frac{\partial Tyz}{\partial \zeta}\right) + \xi_{,z}^m\left(\frac{\partial Tzx}{\partial \xi} + \frac{\partial Tzy}{\partial \eta} + \frac{\partial Tzz}{\partial \zeta}\right)\right], \tag{44}$$

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where

$$T_{xx} = J(\xi_{,x}\sigma_{xx} + \xi_{,y}\sigma_{xy} + \xi_{,z}\sigma_{xz}),$$
$$T_{xy} = J(\eta_{,x}\sigma_{xx} + \eta_{,y}\sigma_{xy} + \eta_{,z}\sigma_{xz}),$$
$$T_{xz} = J(\zeta_{,x}\sigma_{xx} + \zeta_{,y}\sigma_{xy} + \zeta_{,z}\sigma_{xz}), \tag{45a}$$

$$T_{yx} = J(\xi_{,x}\sigma_{xy} + \xi_{,y}\sigma_{yy} + \xi_{,z}\sigma_{yz}),$$
$$T_{yy} = J(\eta_{,x}\sigma_{xy} + \eta_{,y}\sigma_{yy} + \eta_{,z}\sigma_{yz}),$$
$$T_{yz} = J(\zeta_{,x}\sigma_{xy} + \zeta_{,y}\sigma_{yy} + \zeta_{,z}\sigma_{yz}), \tag{45b}$$

$$T_{zx} = J(\xi_{,x}\sigma_{xz} + \xi_{,y}\sigma_{yz} + \xi_{,z}\sigma_{zz}),$$
$$T_{zy} = J(\eta_{,x}\sigma_{xz} + \eta_{,y}\sigma_{yz} + \eta_{,z}\sigma_{zz}),$$
$$T_{zz} = J(\zeta_{,x}\sigma_{xz} + \zeta_{,y}\sigma_{yz} + \zeta_{,z}\sigma_{zz}). \tag{45c}$$

In the same manner, the operators in directions $\mathbf{n}_2$ and $\mathbf{n}_3$ (eq. 17) can be written, respectively, as follows:

$$\nabla_2^m \cdot \boldsymbol{\sigma}|_k = \frac{\xi_{,n}^k}{g_{nn}J}\left[\xi_{,n}^x\left(\frac{\partial Txx}{\partial\xi} + \frac{\partial Txy}{\partial\eta} + \frac{\partial Txz}{\partial\zeta}\right) + \xi_{,n}^y\left(\frac{\partial Tyx}{\partial\xi} + \frac{\partial Tyy}{\partial\eta} + \frac{\partial Tyz}{\partial\zeta}\right) + \xi_{,n}^z\left(\frac{\partial Tzx}{\partial\xi} + \frac{\partial Tzy}{\partial\eta} + \frac{\partial Tzz}{\partial\zeta}\right)\right], \tag{46}$$

and

$$\nabla_3^m \cdot \boldsymbol{\sigma}|_k = \frac{1}{J}n_3^k\left[n_3^x\left(\frac{\partial Txx}{\partial\xi} + \frac{\partial Txy}{\partial\eta} + \frac{\partial Txz}{\partial\zeta}\right) + n_3^y\left(\frac{\partial Tyx}{\partial\xi} + \frac{\partial Tyy}{\partial\eta} + \frac{\partial Tyz}{\partial\zeta}\right) + n_3^z\left(\frac{\partial Tzx}{\partial\xi} + \frac{\partial Tzy}{\partial\eta} + \frac{\partial Tzz}{\partial\zeta}\right)\right], \tag{47}$$

where n_3^k is the kth value of vector $\mathbf{n}_3$.

4.2 ADE CFS-MPML for the stress–velocity relation at the free surface

From the free surface boundary condition (eq. 41) and stress-velocity relation (eqs 2 and 3), the velocity derivatives satisfy the following equation at the free surface:

$$\mathbf{D}_\zeta\begin{pmatrix} v_{x,\zeta} \\ v_{y,\zeta} \\ v_{z,\zeta}\end{pmatrix} = -\mathbf{D}_\xi\begin{pmatrix} v_{x,\xi} \\ v_{y,\xi} \\ v_{z,\xi}\end{pmatrix} - \mathbf{D}_\eta\begin{pmatrix} v_{x,\eta} \\ v_{y,\eta} \\ v_{z,\eta}\end{pmatrix}, \tag{48}$$

where the coefficient matrices are

$$\mathbf{D}_\zeta = \begin{pmatrix} (\lambda+2\mu)\zeta_{,x}\zeta_{,x} + \mu(\zeta_{,y}\zeta_{,y} + \zeta_{,z}\zeta_{,z}) & (\lambda+\mu)\zeta_{,x}\zeta_{,y} & (\lambda+\mu)\zeta_{,x}\zeta_{,z} \\ (\lambda+\mu)\zeta_{,x}\zeta_{,y} & (\lambda+2\mu)\zeta_{,y}\zeta_{,y} + \mu(\zeta_{,x}\zeta_{,x} + \zeta_{,z}\zeta_{,z}) & (\lambda+\mu)\zeta_{,y}\zeta_{,z} \\ (\lambda+\mu)\zeta_{,x}\zeta_{,z} & (\lambda+\mu)\zeta_{,y}\zeta_{,z} & (\lambda+2\mu)\zeta_{,z}\zeta_{,z} + \mu(\zeta_{,x}\zeta_{,x} + \zeta_{,y}\zeta_{,y})\end{pmatrix}, \tag{49}$$

$$\mathbf{D}_\xi = \begin{pmatrix} (\lambda+2\mu)\zeta_{,x}\xi_{,x} + \mu(\zeta_{,y}\xi_{,y} + \zeta_{,z}\xi_{,z}) & \lambda\zeta_{,x}\xi_{,y} + \mu\zeta_{,y}\xi_{,x} & \lambda\zeta_{,x}\xi_{,z} + \mu\zeta_{,z}\xi_{,x} \\ \lambda\zeta_{,y}\xi_{,x} + \mu\zeta_{,x}\xi_{,y} & (\lambda+2\mu)\zeta_{,y}\xi_{,y} + \mu(\zeta_{,x}\xi_{,x} + \zeta_{,z}\xi_{,z}) & \lambda\zeta_{,y}\xi_{,z} + \mu\zeta_{,z}\xi_{,y} \\ \lambda\zeta_{,z}\xi_{,x} + \mu\zeta_{,x}\xi_{,z} & \lambda\zeta_{,z}\xi_{,y} + \mu\zeta_{,y}\xi_{,z} & (\lambda+2\mu)\zeta_{,z}\xi_{,z} + \mu(\zeta_{,x}\xi_{,x} + \zeta_{,y}\xi_{,y})\end{pmatrix}, \tag{50}$$

$$\mathbf{D}_\eta = \begin{pmatrix} (\lambda+2\mu)\zeta_{,x}\eta_{,x} + \mu(\zeta_{,y}\eta_{,y} + \zeta_{,z}\eta_{,z}) & \lambda\zeta_{,x}\eta_{,y} + \mu\zeta_{,y}\eta_{,x} & \lambda\zeta_{,x}\eta_{,z} + \mu\zeta_{,z}\eta_{,x} \\ \lambda\zeta_{,y}\eta_{,x} + \mu\zeta_{,x}\eta_{,y} & (\lambda+2\mu)\zeta_{,y}\eta_{,y} + \mu(\zeta_{,x}\eta_{,x} + \zeta_{,z}\eta_{,z}) & \lambda\zeta_{,y}\eta_{,z} + \mu\zeta_{,z}\eta_{,y} \\ \lambda\zeta_{,z}\eta_{,x} + \mu\zeta_{,x}\eta_{,z} & \lambda\zeta_{,z}\eta_{,y} + \mu\zeta_{,y}\eta_{,z} & (\lambda+2\mu)\zeta_{,z}\eta_{,z} + \mu(\zeta_{,x}\eta_{,x} + \zeta_{,y}\eta_{,y})\end{pmatrix}. \tag{51}$$

Thus at the free surface, additional derivative terms with respect to ξ and η will appear in eq. (9) due to eq. (48). These additional derivative terms have to be taken into account to derive the PML equations at the free surface, otherwise the simulation will become unstable or have large reflections (Zhang & Shen 2010). For a flat surface in the Cartesian coordinates, Zhang & Shen (2010) showed how to include these additional derivative terms in the PML equations. Alternatively, we can express eq. (48) in terms of the strain components in the local Cartesian coordinates that the z'-axis points to the normal direction $\mathbf{n}_3$ of the free surface and x'-axis aligns with the PML boundary as

$$\varepsilon_{x'z'} = 0, \tag{52a}$$

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$$\varepsilon_{y'z'} = 0, \tag{52b}$$

$$(\lambda + 2\mu)\varepsilon_{z'z'} = -\lambda(\varepsilon_{x'x'} + \varepsilon_{y'y'}). \tag{52c}$$

By this equation, we only need to derive the PML equations for $\varepsilon_{x'x'}$, $\varepsilon_{y'y'}$ and $\varepsilon_{x'y'}$, and the effect of the additional derivative terms on the PML equations for $\varepsilon_{x'z'}$, $\varepsilon_{y'z'}$ and $\varepsilon_{z'z'}$ will be automatically incorporated.

Similar to the PML equations of Hooke's law in the Cartesian coordinates (eqs 36 and 37), the governing equations of Hooke's law in the local Cartesian coordinates with PML auxiliary variables are expressed as follows:

$$\varepsilon_{x'x',t} = \frac{1}{\beta_{x'}} v_{x',x'} - \frac{1}{\beta_{x'}} V_{x'}^{x'}, \tag{53}$$

$$\varepsilon_{y'y',t} = \frac{1}{\beta_{y'}} v_{y',y'} - \frac{1}{\beta_{y'}} V_{y'}^{y'}, \tag{54}$$

$$\varepsilon_{x'y',t} = \frac{1}{2}\left(\frac{1}{\beta_{x'}} v_{y',x'} - \frac{1}{\beta_{x'}} V_{y'}^{x'}\right) + \frac{1}{2}\left(\frac{1}{\beta_{y'}} v_{x',y'} - \frac{1}{\beta_{y'}} V_{x'}^{y'}\right), \tag{55}$$

where

$$V_{i',t}^{p'} + \left(\alpha_{p'} + \frac{d_{p'}}{\beta_{p'}}\right) V_{i'}^{p'} = \frac{d_{p'}}{\beta_{p'}} v_{i',p'}. \tag{56}$$

The PML equations in eqs (53)–(56) are only for main damping direction. Equations for the two auxiliary damping directions could be derived in the similar way and omitted here.

To solve eqs (53)–(56), we need to replace the derivatives with respect to the local Cartesian coordinates with those with respect to the curvilinear coordinates. Define the coordinate mapping matrix between the local Cartesian coordinates (i', j', k') and global Cartesian coordinates (i, j, k) as

$$\beta_{i'j} = \beta_{ji'} = \mathbf{u}_{i'} \cdot \mathbf{u}_j, \tag{57}$$

where $\mathbf{u}_{i'}$ is the i'th base vector of the local Cartesian coordinates and $\mathbf{u}_j$ is the jth base vector of the global Cartesian coordinates, we can get (Einstein summation is used on the repeated indices)

$$v_{i',j'} = \beta_{i'i}\beta_{j'j} v_{i,j} = \beta_{i'i}\beta_{j'j}\beta_{jq} v_{i,q}, \tag{58}$$

where subscript q is the axis of the curvilinear coordinates. We express eqs (53)–(56) as follows (Einstein summation is used on repeated indices i', j', i, j, p', q):

$$\varepsilon_{x'x',t} = \frac{1}{\beta_{x'}} \beta_{x'i}\beta_{x'j}\beta_{jq} v_{i,q} - \frac{1}{\beta_{x'}} V_{x'}^{x'}, \tag{59}$$

$$\varepsilon_{y'y',t} = \frac{1}{\beta_{y'}} \beta_{y'i}\beta_{y'j}\beta_{jq} v_{i,q} - \frac{1}{\beta_{y'}} V_{y'}^{y'}, \tag{60}$$

$$\varepsilon_{x'y',t} = \frac{1}{2}\left(\frac{1}{\beta_{x'}} \beta_{y'i}\beta_{x'j}\beta_{jq} v_{i,q} - \frac{1}{\beta_{x'}} V_{y'}^{x'}\right) + \frac{1}{2}\left(\frac{1}{\beta_{y'}} \beta_{x'i}\beta_{y'j}\beta_{jq} v_{i,q} - \frac{1}{\beta_{y'}} V_{x'}^{y'}\right), \tag{61}$$

where

$$V_{i',t}^{p'} + \left(\alpha_{p'} + \frac{d_{p'}}{\beta_{p'}}\right) V_{i'}^{p'} = \frac{d_{p'}}{\beta_{p'}} \beta_{i'i}\beta_{p'j}\beta_{jq} v_{i,q}. \tag{62}$$

We can then obtain the complete strain tensor $\varepsilon_{i'j'}$ from eqs (59) to (62) and relation eq. (52c), and get the final strain tensors at the free surface by

$$\varepsilon_{ij} = \beta_{ii'}\beta_{jj'}\varepsilon_{i'j'}. \tag{63}$$

and then stress component σ_{ij}.

5 NUMERICAL EXPERIMENTS

Orthogonal grid is desirable for numerical simulation but may not always be available for complex geometries. However, the non-orthogonality will more or less exist in the curvilinear grid for real applications. Since an orthogonal grid is a special case of the non-orthogonal grid, we can expect that a PML which works on non-orthogonal grids will also work on the orthogonal grids. In this section, we use two numerical

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experiments to illustrate the effects of the ADE CFS-MPML on non-orthogonal grids. In the first numerical experiment, we use a full space model by applying PML at all the directions to compare the performance of different PML models on a curvilinear grid with large grid deformation. In the second numerical experiment, we add free surface at the top of the model to illustrate the importance of the correction of PML due to the free surface boundary condition.

5.1 Full-space model with large grid deformation

In this numerical experiment, we test different PML models (ADE PML, ADE CFS-PML, ADE MPML and ADE CFS-MPML, see Table 1) on a curvilinear grid with large grid deformation (55° between $\mathbf{e}_\xi$ and $\mathbf{e}_\zeta$, see Fig. 3) to compare their performances in terms of stability and absorption. The model represents an infinite homogeneous elastic medium with *P*-wave speed $v_p = 6000$ m s^{-1}, *S*-wave speed $v_s = 3464$ m s^{-1} and density $\rho = 2670$ kg m^{-3} in full space. Parallelogram-shaped gridpoints in one plane are used to discretize the computational volume, thus allowing a 55° angle of deformation comparing the orthogonal gridpoints (90°). We use an explosive source with a Ricker wavelet, of the central frequency 2.5 Hz. At the 6 boundaries of the physical domain, 24 PML layers are used to absorb the outgoing waves.

We take the optimal values of the parameter α, β and d of the CFS-PML following Zhang & Shen (2010). For MPML, we take the auxiliary coefficients as follows:

$$p_{ij} = \left(\frac{x}{L}\right)^2 \cdot p_{\max}, \tag{64}$$

where x is the distance to the PML-interior interface, L is the width of the PML layer and $p_{\max} = 0.3$. Evaluating the auxiliary coefficients as the function of distance to PML-interior interface (eq. 64) allows smaller multi-axial damping profile at the beginning layers of PML and then smaller reflection than usually constant p_{ij}.

Fig. 4 illustrates snapshots of four comparing simulations with the PML models listed in Table 1. From the comparison of the first snapshot of each modelling we can see that when the seismic wave enters the upper absorbing boundary, the wavefield from CFS-PMLs (Case C and D) gets bended (Zhang & Shen 2010), while amplitude smoothly decreasing to zero near the outermost boundary. Meanwhile, the wavefiled derived by PMLs without CFS (Case A and B) still preserves large value near the outermost boundary of PML. In other words, the CFS-PML reduces the seismic wave that propagating parallel with PML by bending wave front, and then increase the absorbing effect, especially at grazing incidence.

In the long-time simulation, standard PML (Case A) solution firstly behaves unstable. When we use CFS, it delays the occurrence of unstable solution. Because a relatively small $p_{\max}$ and the PML distance-varying auxiliary coefficient (eq. 64) are used, the modelling with ADE MPML (Case B) is unstable either. Only the ADE CFS-MPML can get a stable result for the parameters we choose in this simulation. Generally speaking, a larger auxiliary coefficient of MPML will ensure a more stable solution. Unshown results also suggest that, for Case B, a larger $p_{\max}$ or constant p_{ij} ($p_{ij} = p_{\max} = 0.3$) can stabilize the solution. It seems that the larger auxiliary coefficient is required to stabilize the MPML's solution if the CFS is not added. The same phenomenon was also observed by Martin *et al.* (2008). It may be caused by that CFS can more efficiently absorb the near grazing energy by bending wave front to PML (Fig. 4, Case B and D at time 1.8 s). With CFS, we can use smaller $p_{\max}$, and then smaller reflection.

To quantitatively illustrate the stability of PML solutions, the evolution of the total energy in the physical domain during the simulations is investigated. The total energy was calculated at each time step by:

$$E = \frac{1}{2}\int_\Omega \rho ||\mathbf{v}||^2 \mathrm{d}\Omega + \frac{1}{2}\int_\Omega \rho \sigma_{ij}\epsilon_{ij}\mathrm{d}\Omega, \tag{65}$$

where Ω represents the physical domain. Fig. 5 shows the variation of the total energy with respect to the simulation time. The CFS-PMLs (Case C and Case D) have less total energy between the time window of 5–9 s, which indicate more efficient absorption of outgoing waves than the standard PMLs (Case A and Case B), respectively on curvilinear grids.

Table 1. Conditions of the four cases.

Cases	CFS	Multi-axial	Name
A	No	No	ADE PML
B	No	Yes	ADE MPML
C	Yes	No	ADE CFS-PML
D	Yes	Yes	ADE CFS-MPML

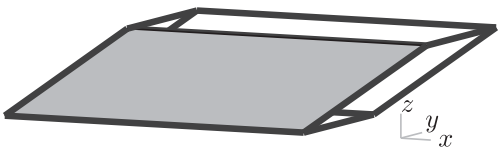


Figure 3. 3-D geometry of model used in this study. The grey plane is at the centre of the computational volume (the other side of the grey plane is not shown for clearance).

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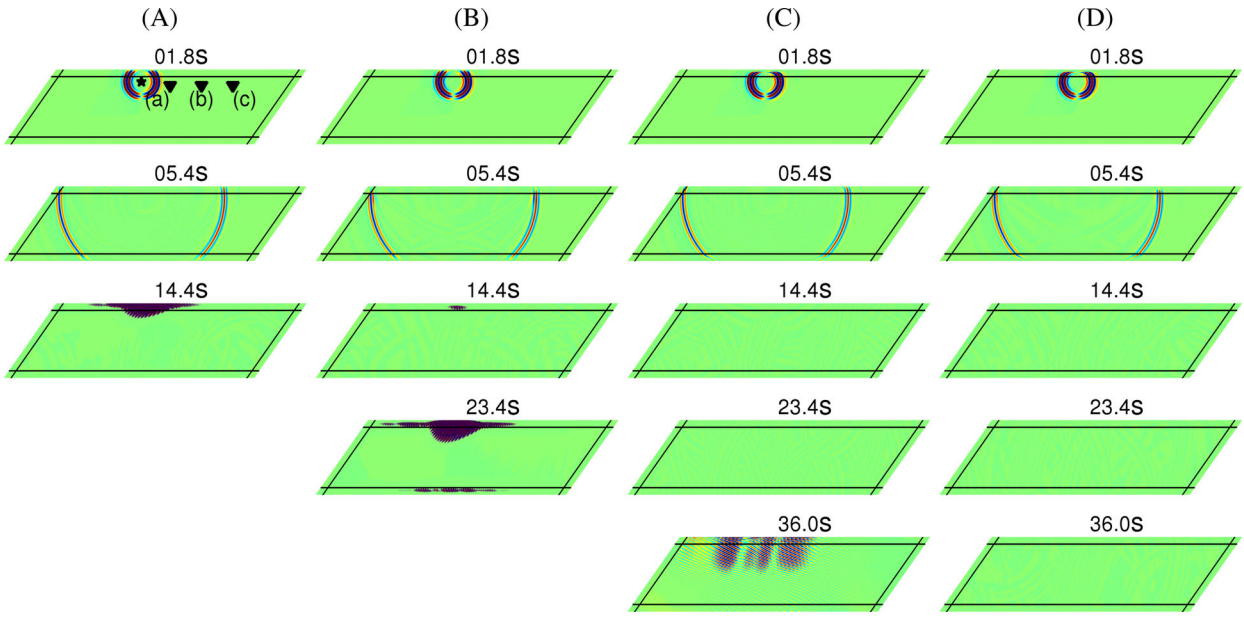


Figure 4. Snapshots of the horizontal component of the particle velocity of the four cases (Table 1) at vertical ξ–ζ plane across the source (Fig. 3). The time of each snapshot is labeled at the top of each snapshot. The black lines in the snapshots denote the separation of PML from the physical domain. The source and the three receivers [point (a), (b) and (c) from the left to the right in order] are represented by a five-point start and inverted triangles, respectively.

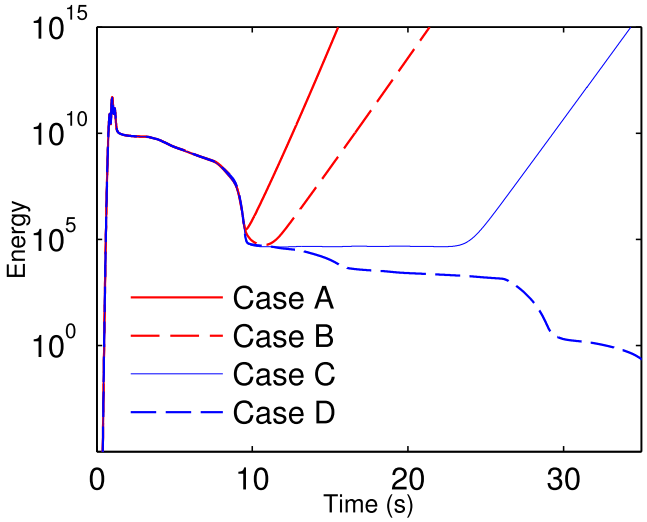


Figure 5. Energy evolutions in the physical domain of the four cases (Table 1).

For more informations about the performance of each PML schemes, we compare the numerical simulations derived by different PML truncation schemes with analytic solution at three selected points (a), (b) and (c) (inverted triangles in Fig. 4). Point (a) has the nearest offset relative to the source, while (c) the farthest offset. The rms error of each PML solution comparing the analytic solution is calculated to evaluate the performance of the PML schemes, and is labeled at the end of the seismogram (Fig. 6). As revealed by the velocity seismogram comparisons (Fig. 6) at the three receivers, Case C has smaller rms errors than Case A, and Case D has smaller rms errors than Case B, especially at the far offset [receiver (c)]. That is because the CFS PML absorbs the grazing incident wave better than the standard PML (Drossaert & Giannopoulos 2007b; Komatitsch & Martin 2007; Zhang & Shen 2010). The multi-axial damping added in Case B makes the rms error a little larger than Case A, so does Case D comparing to Case C. However, Case D preserves stable solution (Fig. 5) as compared to Case C, when the grid deformation exists.

This numerical test illustrates that the ADE CFS-MPML can achieve stable seismic wave simulation on the curvilinear grids with the best trade-off in terms of the absorbing efficiency and stability.

5.2 Half-space model with a free surface

Section 4 discussed how to implement the ADE CFS-MPML scheme when the absorbing layer reaches the free surface. The key point of this implementation is the correction of damping direction at the free surface, because two horizontal derivatives are needed to calculate the

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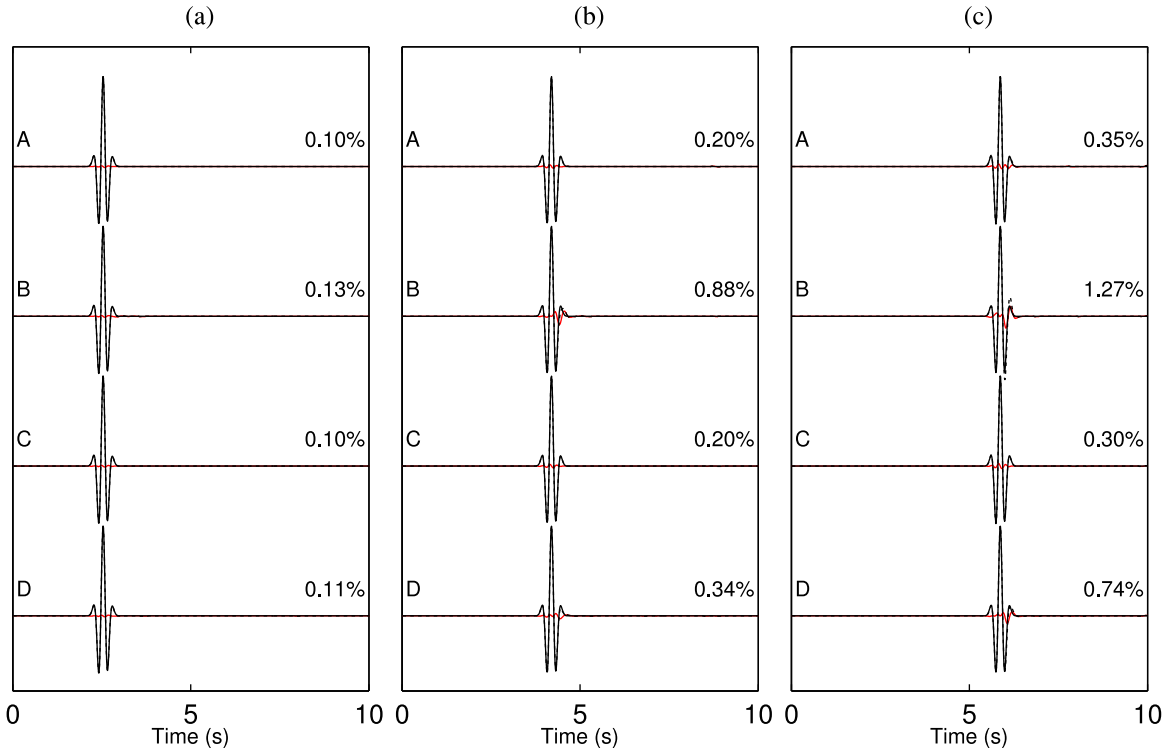


Figure 6. Comparison between numerical simulations derived by different PML schemes (dashed lines) and the analytical solution (solid lines) at the receiver (a), (b) and (c) (marked by the three inverted triangles in Fig. 4). The differences between the PML solutions and the analytical ones are illustrated by red lines. Only the first 10 s results of the modellings are illustrated. The rms error of each case is marked at the end of seismogram.

vertical one. In this section, we validate this correction by modelling the two comparing simulations in the half space, one includes the surface correction at the intersection of PML and the free surface, and the other does not. Same meshing grid as the previous modelling is used in this simulation, except a free-surface condition is implemented. Moreover, to generate strong *S* wave, a double couple source is used. The other parameters are the same as that in the full-space model. The source is located close to the PML and the free surface.

Snapshots (Fig. 7) of the comparing simulations illustrate that the ADE CFS-MPML solution with surface correction (Fig. 7, left-hand column) preserves the stability even long time after the wave vanishes. For the ADE CFS-MPML without surface correction (Fig. 7, right-hand column), the solution goes unstable when the main energy entrances the corner where the free surface and PML intersect. These comparing simulations suggest that the surface correction of PML is required.

6 CONCLUSION

In this paper, we extended the ADE CFS-PML (Zhang & Shen 2010) to general curvilinear (non-orthogonal) grids for elastic wave modelling.

Unlike the common implementations to absorb the waves in the computational space, we applied the damping along the perpendicular direction of the PML layer in the local Cartesian coordinates. Instead of mapping all the wavefield to the local Cartesian coordinates, we directly expressed the perpendicular and parallel components of the gradient operator in the local Cartesian coordinates in terms of the derivatives in the curvilinear coordinates. Such approach leads to complete first-order partial differential equations for PML. It is thus easy to be incorporated with numerical schemes on curvilinear grids.

The elastic wave modelling on curvilinear grids exhibits anisotropic effects in the computational space, which may cause the PML to become amplifying layers as the PML in anisotropic media (Halpern *et al.* 2011). To stabilize PML solution on curvilinear grids, we adapted the multi-axial damping, which was originally proposed by Meza-Fajardo & Papageorgiou (2008) to avoid the PML amplifying in anisotropic media, to derive ADE CFS-MPML.

In the ADE CFS-MPML, the main damping profile is perpendicular to PML surface, while the other two auxiliary profiles are parallel to the PML surface. We derive the ADE CFS-MPML equations for the interior region and for the free surface separately because the free surface boundary condition modifies the elastic wave equations.

We illustrated the stability of the ADE CFS-MPML for finite-difference elastic wave simulations on curvilinear grids by two numerical experiments. The full-space numerical experiment showed that the ADE CFS-MPML with properly selected parameters can get stable simulation on curvilinear grids with large grid deformation. The half-space numerical experiment demonstrated the importance of the PML correction due to the free surface boundary condition to achieve stable simulations.

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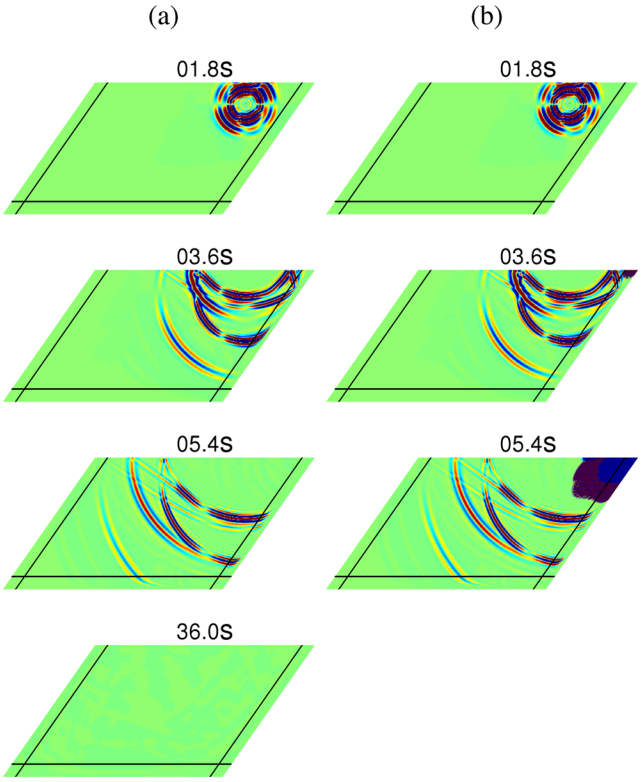


Figure 7. Snapshots of ADE CFS-MPML solution in the half space with (left-hand column, a) and without (right-hand column, b) surface correction. The time of each snapshot is labeled at the top of each snapshot. The black lines in the snapshots denote the boundary of the PML and the physical domain.

We note that all the simulations in this work were performed for linear elastic media. The ADE CFS-MPML could be extended to more general media in future studies.

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